

Advances in Finite Element Methods and Parallel Scientific Computing

November 28 – December 2, 2026

IIT Bhubaneswar

Academic Program

Venue: Room No. L02M, Lalitgiri Hall of Learning

Inauguration Program: 9:00 AM – 9:30 AM, L02M (Lalitgiri Hall of Learning), 28th November 2026

Day 1: November 28, 2026

Time	Speaker / Title of the Talk
	Chair: Prof. Bhupen Deka
9:30 AM – 10:10 AM	Prof. Rajen Kumar Sinha <i>A geometric transformation based finite element method for the elliptic interface problems</i>
10:15 AM – 11:05 AM	Dr. Asha Dond <i>Lecture Series on Adaptive FEM: Convergence and Quasi-optimality – Lecture 1</i>
11:05 AM – 11:35 AM: TEA BREAK	
11:35 AM – 12:25 PM	Dr. Pratibhamoy Das <i>Lecture Series Parallel Computing – Lecture 1</i>
12:30 PM – 12:50 PM	Contributed Talk: Rohit Kumar Yadav <i>A parameter uniform Numerical method for Singularly Perturbed Semilinear Convection diffusion problems with discontinuous source term</i>
1:00 PM – 2:30 PM: LUNCH BREAK	
	Chair: Dr. Pratibhamoy Das
2:30 PM – 3:10 PM	Dr. Sangita Yadav <i>Hybridizable Discontinuous Galerkin (HDG) Method - 1</i>
3:15 PM – 3:35 PM	Contributed Talk: Ajay Kumar <i>Hybrid high-order methods for sixth order problems with a mixed formulation</i>
3:40 PM – 4:00 PM	Contributed Talk: Suraj Kumar <i>Parameter-Uniform Convergence and Superconvergence of NIPG Method for Degenerate Singularly Perturbed Convection-Diffusion Problems</i>
4:00 PM – 4:25 PM: TEA BREAK	
4:25 PM – 4:45 PM	Contributed Talk: Dilip Sarkar <i>A Finite Element Framework for Domain Decomposition with Non-Matching Grids: A Nitsche-Based Approach</i>
4:50 PM – 5:10 PM	Contributed Talk: Kumar Mritunjay <i>Boundary Feedback Stabilization and Finite Element Approximation of a Strongly Damped Semilinear Wave Equation</i>

Time	Speaker / Title of the Talk
	Chair: Dr. Bankim Chandra Mandal
9:30 AM – 10:20 AM	Dr. Pratibhamoy Das <i>Lecture Series on Parallel Computing - 2</i>
10:25 AM – 11:05 AM	Dr. Saumya Bajpai <i>A Three-Step Two-Grid DG Method for the Kelvin-Voigt Visco-elastic Flow Model</i>
11:05 AM – 11:35 AM: TEA BREAK	
11:35 AM – 12:15 PM	Prof. Neela Natarajan <i>Unified numerical analysis for thermoelastic diffusion and thermo-poroelasticity of thin plates</i>
12:20 PM – 1:00 PM	Dr. Sangita Yadav <i>Hybridizable Discontinuous Galerkin (HDG) Method - 2</i>
1:00 PM – 2:30 PM: LUNCH BREAK	
	Chair: Dr. Sangita Yadav
2:30 PM – 3:20 PM	Dr. Arbaz Khan <i>Lecture Series on Mixed FEM for Multiphysics Problems - 1</i>
3:25 PM – 3:45 PM	Contributed Talk: Jitendra Nath Naik <i>Optimal Error Estimates of a Finite Element Method for Semilinear SPDEs with Non-smooth Initial Data</i>
3:50 PM – 4:15 PM: TEA BREAK	
4:20 PM – 4:40 PM	Contributed Talk: Arijit Pal <i>Dimensional-Splitting Weak Galerkin FEM for Singularly Perturbed Initial-Boundary Value Problems</i>
4:40 PM – 5:00 PM	Contributed Talk: Dr. Rashmita Baruah <i>A Weak Galerkin Method with Forward Euler Time Discretization for Singularly Perturbed Parabolic Interface Problems</i>
5:00 PM – 5:20 PM	Contributed Talk: Pratyay Mondal <i>Stability and convergence of a decoupled discontinuous Galerkin scheme for the Oldroyd Model of order one</i>

Time	Speaker / Title of the Talk
	Chair: Prof. Rajen Sinha
9:30 AM – 10:20 AM	Dr. Asha Dond <i>Lecture Series on Adaptive FEM: Convergence and Quasi-optimality – Lecture 2</i>
10:25 AM – 11:05 AM	Prof. Rajendra K. Ray <i>New Higher-Order Super-Compact Schemes for Three-Dimensional Fluid Flow and Heat Transfer in Non-Newtonian Fluids</i>
11:05 AM – 11:25 AM: TEA BREAK	
11:25 AM – 12:15 PM	Prof. Bhupen Deka <i>Lecture series on Weak Galerkin Solver for Elliptic BVP on Polygonal Meshes – 1</i>
12:20 PM – 1:00 PM	Dr. Y. Bhumkar <i>Scientific Computing of Computational Aero Acoustic Problems</i>
1:00 PM – 2:30 PM: CONFERENCE LUNCH	
3:00 PM – 7:30 PM	Free Time/Local Excursion

AFEMPSC

IIT BHUBANESWAR

Time	Speaker / Title of the Talk
	Chair: Prof. Rajendra K. Ray
9:30 AM – 10:20 AM	Dr. Arbaz Khan <i>Lecture Series on Mixed FEM for Multiphysics Problems – Lecture 2</i>
10:25 AM – 11:05 AM	Dr. Abhinav Jha <i>Finite Element Methods for Convection-Dominated Problems: From Galerkin to Stabilization</i>
11:05 AM – 11:15 AM: TEA BREAK	
11:20 AM – 12:00 PM	Dr. Nagaiah Chamakuri <i>Computational Challenges in Cardiac Electromechanics: Robust Algorithms and Scalable Solvers</i>
12:05 PM – 12:25 PM	Contributed Talk: Deeksha Tomer <i>Domain Decomposition Methods for PDEs with Time Delay</i>
12:25 PM – 1:05 PM	Dr. Rakesh Kumar <i>Divergence-free reconstruction of 2-D vector fields: Application to induction equation</i>
1:10 PM – 2:30 PM: LUNCH BREAK	
	Chair: Dr. Sanjib Acharya
2:30 PM – 3:20 PM	Prof. Bhupen Deka <i>Lecture series on Weak Galerkin Solver for Elliptic BVP on Polygonal Meshes - 2</i>
3:25 PM – 3:45 PM	Contributed Talk: Monirul Islam <i>Unconditionally Energy Stable Ultra-Weak Discontinuous Galerkin Scheme with Generalized Numerical Fluxes for the Allen-Cahn Equation</i>
3:50 PM – 4:10 PM	Contributed Talk: Dr. Ritesh Kumar Singh <i>Exact Solution of the Riemann Problem in a Perfect Gas with Magnetic Field Using Lie Symmetry Analysis</i>
4:10 PM – 4:30 PM: TEA BREAK	
4:30 PM – 4:50 PM	Contributed Talk: Ashish Kumar Saini <i>Enhancing thermal performance in square enclosures filled with oxide nano-fluids subjected to magnetic fields: a numerical study</i>
4:55 PM – 5:15 PM	Contributed Talk: Dr. Harish Malleshham Nagula <i>Vectorized Mortar Primal Hybrid Finite Element Method</i>

Time	Speaker / Title of the Talk
	Chair: Dr. Nagaiah Chamakuri
9:30 AM – 10:20 AM	Prof. Sarvesh Kumar <i>Virtual Element Methods: Recent Developments and Applications</i>
10:25 AM – 11:05 AM	Dr. Surya Maharana <i>On Applications of FEM for the Study of Chemo-hydrodynamic Instabilities</i>
11:05 AM – 11:30 AM: TEA BREAK	
11:30 AM – 11:50 AM	Contributed Talk: Dr. Achyuta Ranjan Dutta Mohapatra <i>Energy norm error estimates of a hybrid high-order method for the linear parabolic integro-differential equations on general meshes</i>
11:55 AM – 12:15 PM	Contributed Talk: Anupama Ghorai <i>A high-order finite element method for multi-dimensional population balance equations.</i>
12:20 PM – 12:40 PM	Contributed Talk: Ayush Agrawal <i>Nitsche method for the Stokes-Poisson-Boltzmann equation with Navier slip boundary condition.</i>
12:45 PM – 1:05 PM	Contributed Talk: Dr. Arya Prakash Padhi <i>Scalable One-Shot 3D Topology Optimization with Multigrid PCG and GPU-Accelerated Finite Element Analysis</i>
1:00 PM – 3:00 PM: LUNCH BREAK	
	Chair: Dr. Arbaz Khan
3:00 PM – 3:20 PM	Contributed Talk: Dr. Ruma Rani Maity <i>An Energy-Decreasing Algorithm for the Finite Element Approximation of Ferronematic Equilibrium States</i>
3:25 PM – 3:45 PM	Contributed Talk: Dr. Shubhangi Dwivedi <i>Managing Connected Ecological Patches Under Species Dispersal Using Optimal Control</i>
3:50 PM – 4:30 PM: CLOSING CEREMONY	

— IIT BHUBANESWAR —